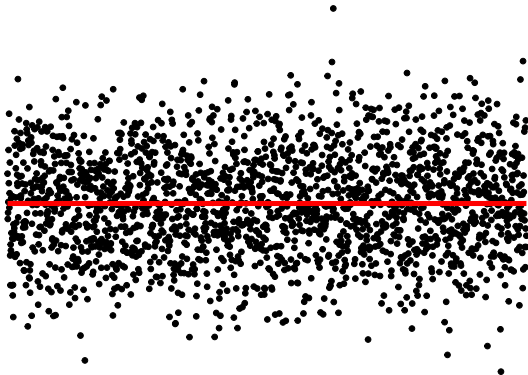


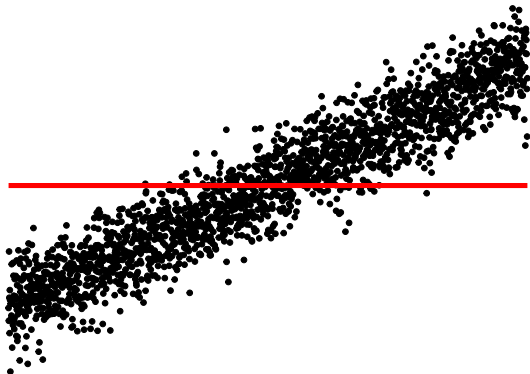
Generalised mixed effects models for changepoint analysis of biomedical time series

Rebecca Killick(r.killick@lancs.ac.uk)

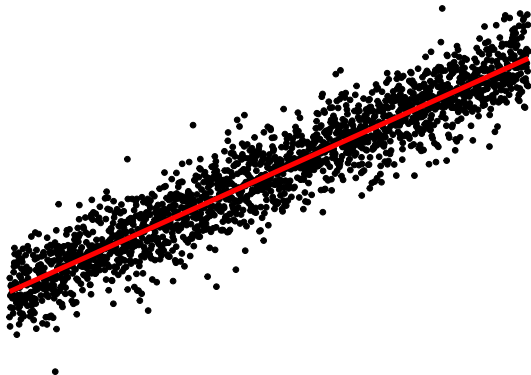
Joint work with Mark Fiecas and Kathryn Cullen (Minnesota)

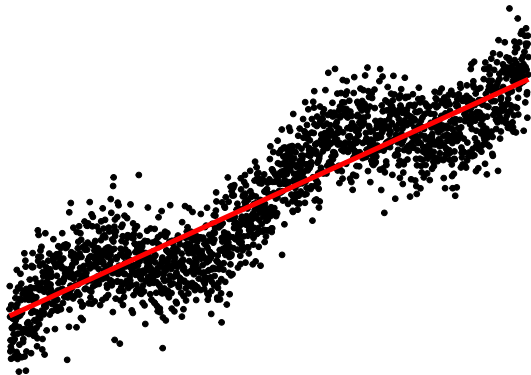
UCLA Seminar Apr 2025



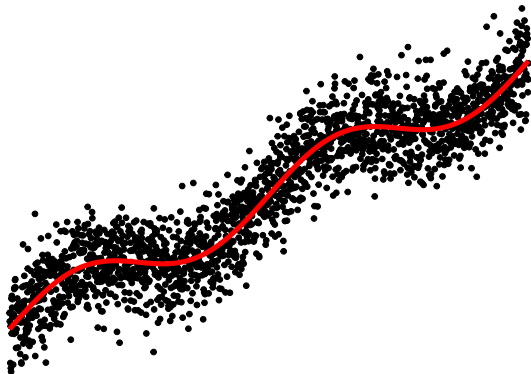


Model development

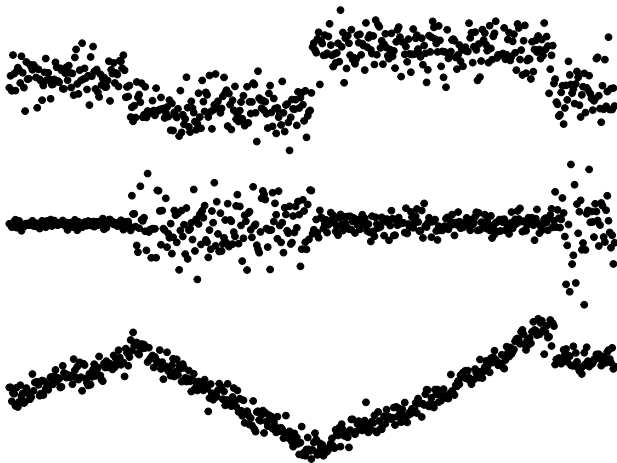




Model development



Model development



- Motivation from Biomedical Time Series
- Background
 - What are changepoints?
 - How do we fit changepoint models?
- A generalised mixed effects model with changepoints
- Simulations + Data Applications
- Summary

Brain Imaging Development of Girls' Emotion and Self (BRIDGES) Study

- Resting state data from 137 girls, no NSSI to severe NSSI.
- 56 spatial locations, 4 regions of interest
- 904 observations from each individual

The Howz Smart System:

- Discreet passive sensors.
- Measure activity in the home.
- Learn about your daily routine.

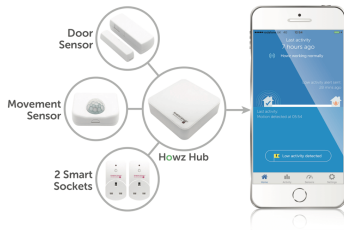
Objective:

- Identify abnormal behaviour:
- Indicate a decline in health or well-being in an older person.

Data:

- 4 people
- Binary yes/no activity in 15 mins
- 8 weeks (56 days) of activity data

howz



- Multiple people
- Repeated observations per person
- Common structure with random variation across people
- Potential changes over time

Commonly used model = random effects model

$$\mathbb{E}(y_{i,j}) = \mu + \mu_i + \epsilon_{i,j}$$

Generalized Linear Mixed Effects Regression (GLMER) model.

$$y_i | \mathbf{b} \sim \text{Distr}(\mu_i, \sigma^2) \quad (1)$$

$$g(\mu) = \mathbf{X}\beta + \mathbf{Z}\mathbf{b}, \quad (2)$$

- $\mathbf{X}$ - external regressors with coefficients β
- Distr - conditional distribution
- $\mathbf{Z}$ - random effects with coefficients $\mathbf{b}$

Standard GLMER estimators work well.

$$y_i|b \sim \text{Distr}\left(0, \sigma_i^2\right) \quad (3)$$

(4)

We want to write the covariance in a similar random effects form.

For Uniform-block (UB) structure we get

$$\Sigma = \tilde{\Sigma}_{\epsilon} \circ \mathbf{Id}(\mathbf{k}) + \Sigma_{\mu} \circ \mathbf{J}(\mathbf{k}). \quad (5)$$

where:

- $\tilde{\Sigma}_{\epsilon} = \text{diag}(\sigma_{\epsilon,1}^2, \dots, \sigma_{\epsilon,K}^2)$
- $\mathbf{Id}(\mathbf{k}) = \text{Bdiag}(\text{Id}_{k_1}, \dots, \text{Id}_{k_K})$, where $\text{Bdiag}(\cdot)$ constructs a block-diagonal matrix
- $\Sigma_{\mu} = (b_{ij})$ be a $K \times K$ covariance matrix
- $\mathbf{J}(\mathbf{k}) = (\mathbf{1}_{k_u \times k_v})$
- $\circ$ be the block Hadamard product

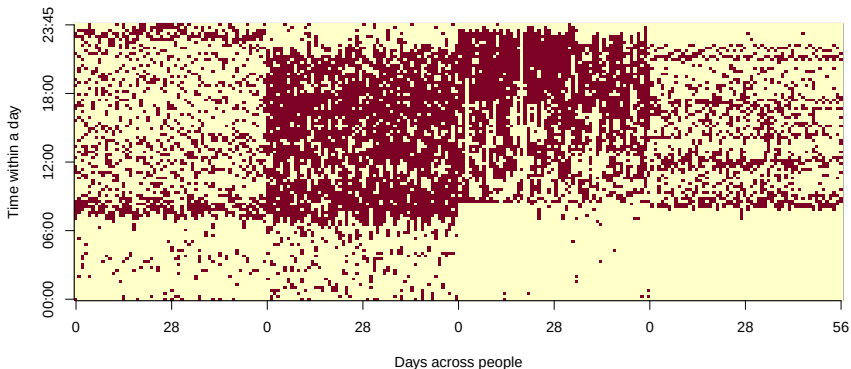
For Heterogeneous-block (HB) structure we get

$$\Sigma = \tilde{\Sigma}_{\epsilon} + \Sigma_{\mu} \circ \mathbf{J}(\mathbf{k}). \quad (6)$$

- May be prohibitive for high dimensional (n relative to K)

Estimate parameters by block-wise method of moments (see ArXiV paper).

What now?

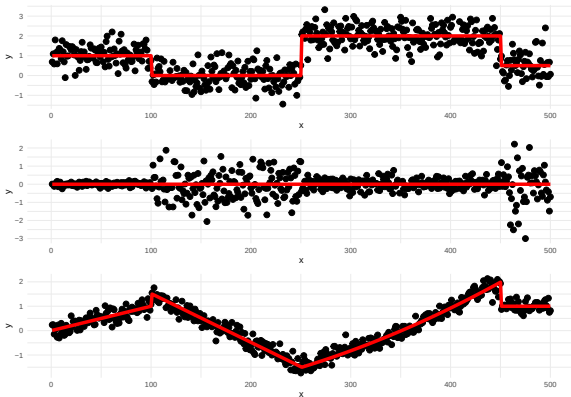


What are changepoints?



For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

There are many different types of change.

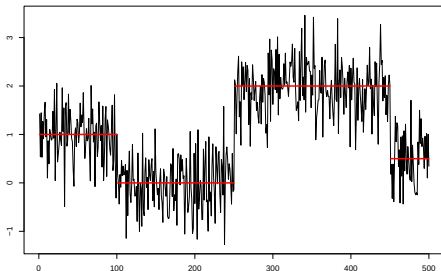




Assume we have time-series data where

$$Y_t | \theta_t \sim N(\theta_t, 1),$$

but where the means, θ_t , are piecewise constant through time.



We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau} \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β . If we infer a changepoint its position is estimated as

$$\tau = \arg \max \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$

This can test can be repeatedly applied to new segments to find multiple changepoints.

Define m to be the number of changepoints, with positions $\tau = (\tau_0, \tau_1, \dots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

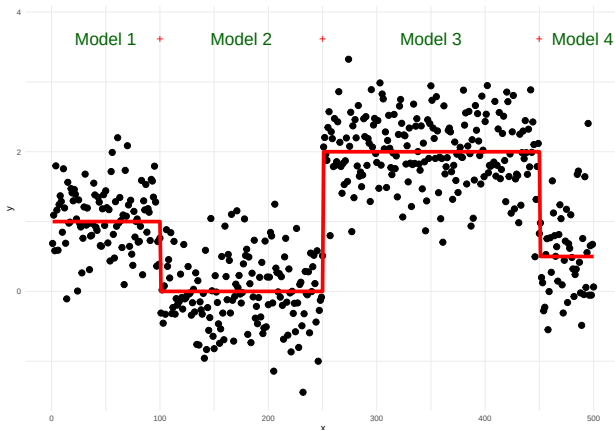
Then one application of the Likelihood ratio test can be viewed as

$$\min_{m \in \{0, 1\}, \tau} \left\{ \sum_{i=1}^{m+1} [-\ell(y_{\tau_{i-1}:\tau_i})] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m, \tau} \left\{ \sum_{i=1}^{m+1} [-\ell(y_{\tau_{i-1}:\tau_i})] + \beta m \right\}$$

Problem

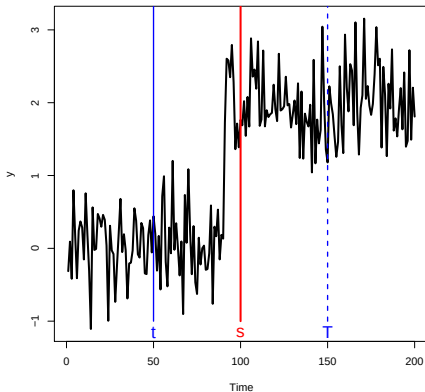


- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell



- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.



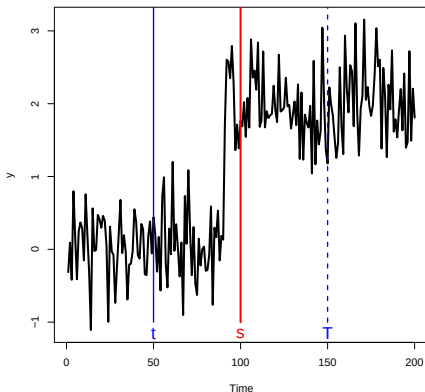


Let $0 < t < s < T$, if

$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

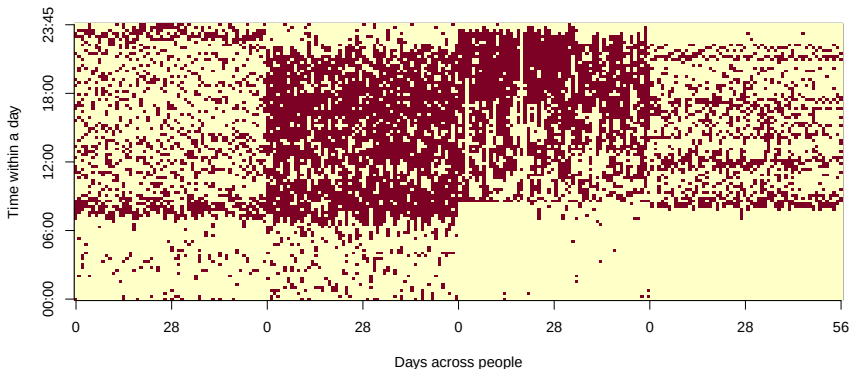
then at any future time $T > s$, t can never be the optimal last changepoint prior to T .

We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.





- Use the GLMER or LMEC likelihood as the cost function for PELT



- What can change?
- Minimum segment length - theoretical minimum? Covariance estimation is hard!
- Penalty for changepoint estimation - BIC
- Assume known group structure

- $n=1000$
- $K=4$ groups, 5 series per group
- Simulate covariances from Wishart distribution and manipulate

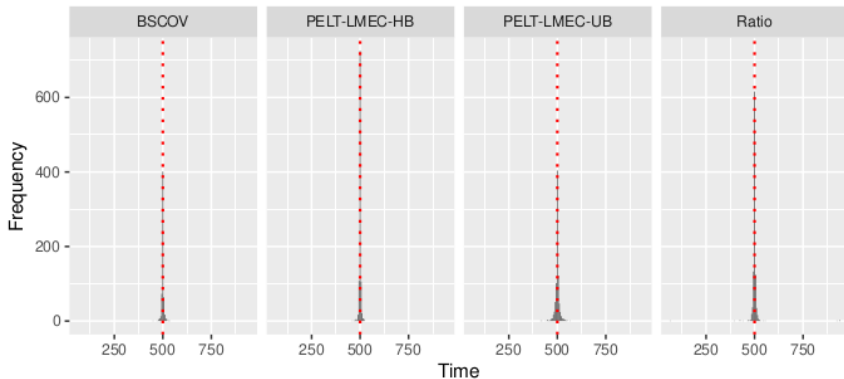
Compare to

- Ratio method (Ryan and Killick (2023)), no group structure
- Wild Sparsified Binary Segmentation (Li et al.(2023)), factor model

Single cpt $p=20$



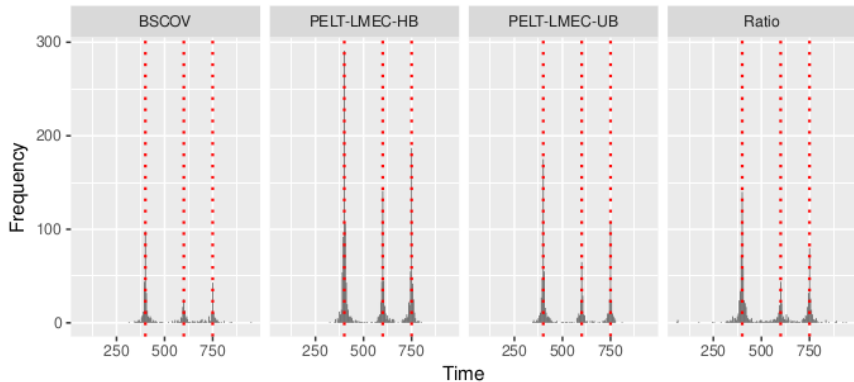
(a) Single changepoint, $p = 20$



3 cpts $p=20$



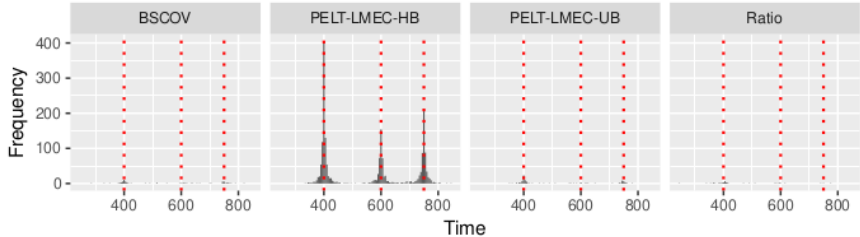
(b) Multiple changepoints, $p = 20$



3 cpts $p=60$

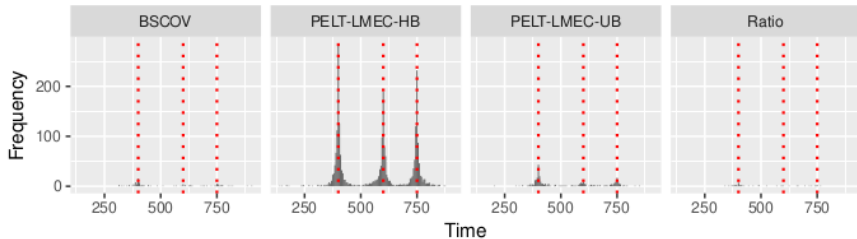


(a) Multiple changepoints, $p = 60$



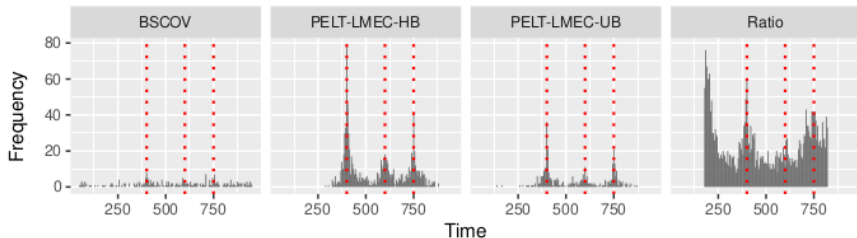


(b) Multiple changepoints, misspecified groups, $p = 60$





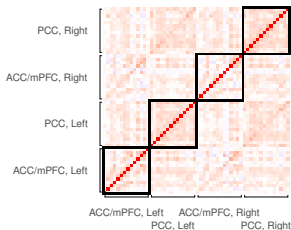
(c) Multiple changepoints, autocorrelated data, $p = 60$



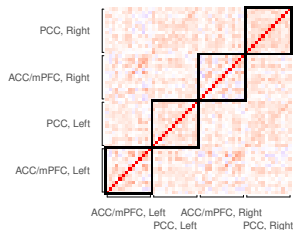
- LMEC-HB best overall
- Mild misspecification doesn't affect cpt detection
- No accounting for autocorrelation is bad



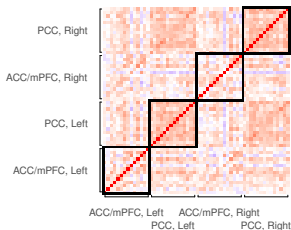
Static FC



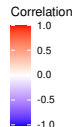
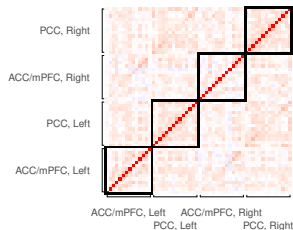
Dynamic FC, Segment 1

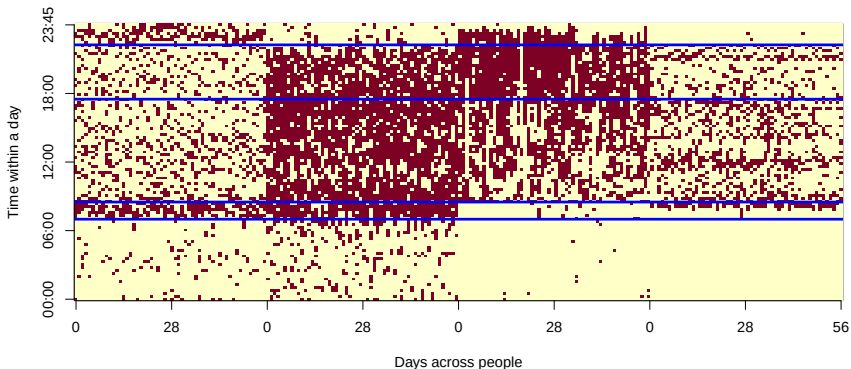


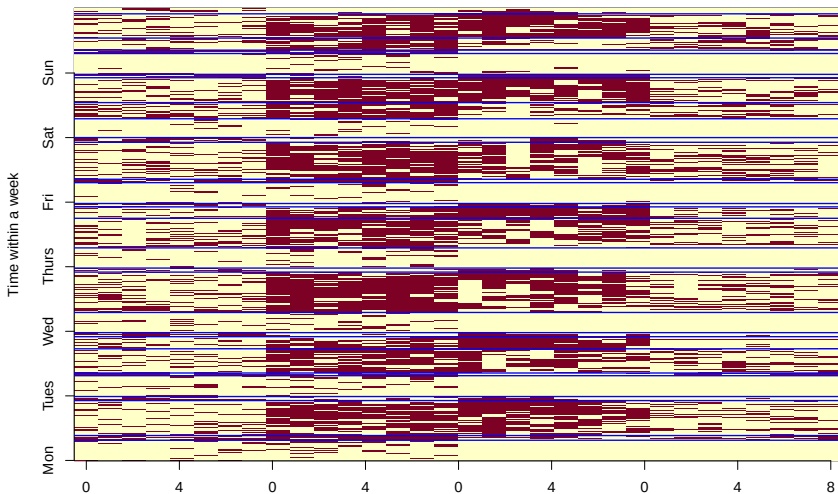
Dynamic FC, Segment 2



Dynamic FC, Segment 3







- Changepoints occur in most data
- Detecting and documenting them improves analyses
- Presented a generalised random effects model with changepoints
- Showed its utility in two very different data settings
- I enjoy working with messy real life data ...
- ... as often it sparks my next research challenge.



- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

